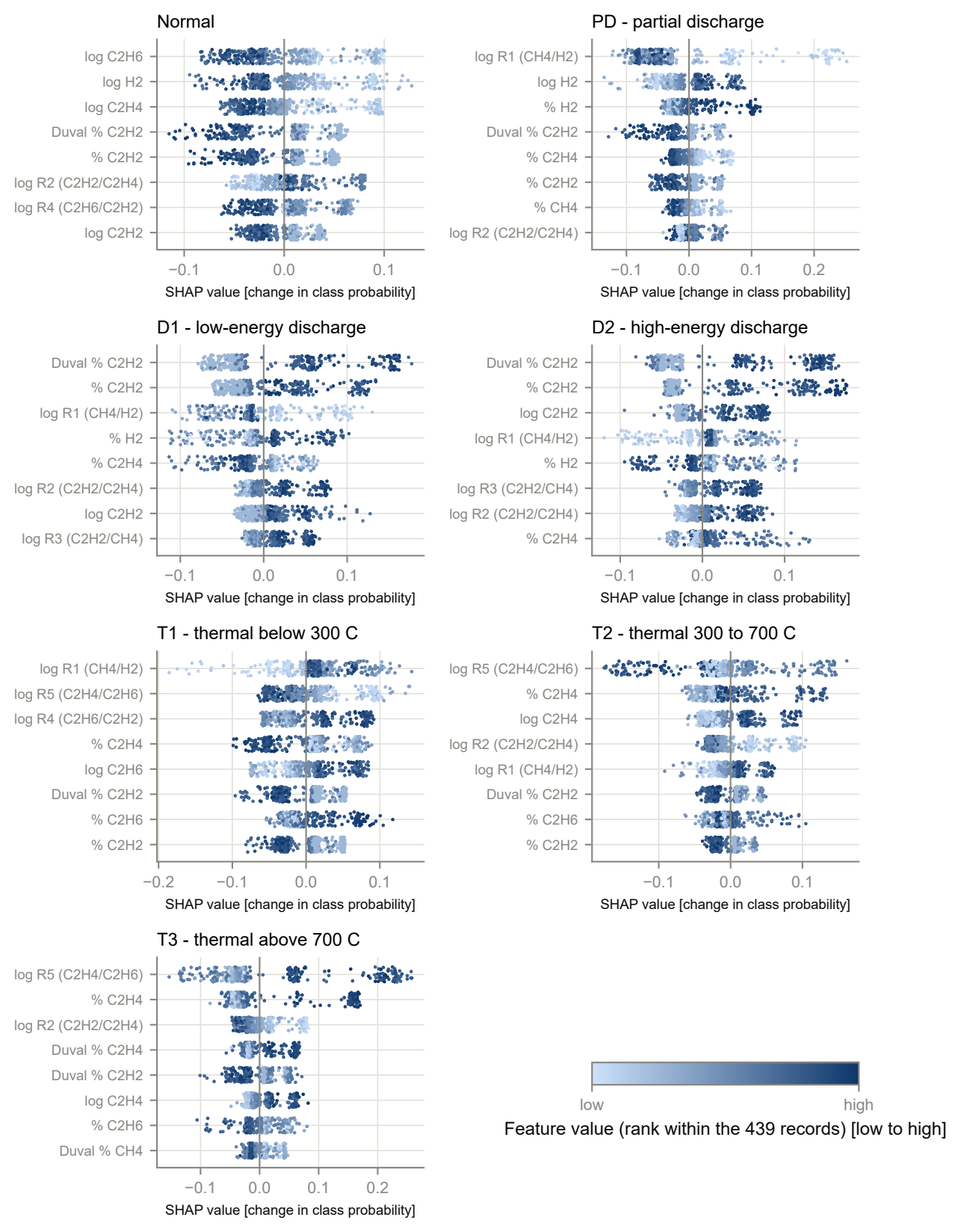

Normal
log C2H6
log H2
log C2H4
Duval % C2H2
% C2H2
log R2 (C2H2/C2H4)
log R4 (C2H6/C2H2)
log C2H2
−0.1
0.0
0.1
SHAP value [change in class probability]
PD - partial discharge
log R1 (CH4/H2)
log H2
% H2
Duval % C2H2
% C2H4
% C2H2
% CH4
log R2 (C2H2/C2H4)
−0.1
0.0
0.1
0.2
SHAP value [change in class probability]
D1 - low-energy discharge
Duval % C2H2
% C2H2
log R1 (CH4/H2)
% H2
% C2H4
log R2 (C2H2/C2H4)
log C2H2
log R3 (C2H2/CH4)
−0.1
0.0
0.1
SHAP value [change in class probability]
D2 - high-energy discharge
Duval % C2H2
% C2H2
log C2H2
log R1 (CH4/H2)
% H2
log R3 (C2H2/CH4)
log R2 (C2H2/C2H4)
% C2H4
−0.1
0.0
0.1
SHAP value [change in class probability]
T1 - thermal below 300 C
log R1 (CH4/H2)
log R5 (C2H4/C2H6)
log R4 (C2H6/C2H2)
% C2H4
log C2H6
Duval % C2H2
% C2H6
% C2H2
−0.2
−0.1
0.0
0.1
SHAP value [change in class probability]
T2 - thermal 300 to 700 C
log R5 (C2H4/C2H6)
% C2H4
log C2H4
log R2 (C2H2/C2H4)
log R1 (CH4/H2)
Duval % C2H2
% C2H6
% C2H2
−0.1
0.0
0.1
SHAP value [change in class probability]
T3 - thermal above 700 C
log R5 (C2H4/C2H6)
% C2H4
log R2 (C2H2/C2H4)
Duval % C2H4
Duval % C2H2
log C2H4
% C2H6
Duval % CH4
−0.1
0.0
0.1
0.2
SHAP value [change in class probability]
low
high
Feature value (rank within the 439 records) [low to high]